

Question

The incorrect proposition number among the following statements about genetic translation is:

Proposition 1:

Each amino acid is encoded by a codon (a segment consisting of three nucleotide residues in mRNA). Among all codons, the total number of codons that have or also possess special signal functions is consistent with the number of amino acid types that exhibit the highest codon degeneracy.

Proposition 2:

If the sequence 5'-AUG UCU UAA-3' is translated in an in vitro system lacking translation release factors, the ribosome will recognize the UAA codon and recruit a tRNA carrying leucine (L), thereby yielding the polypeptide M-S-L.

Proposition 3:

The molar molecular mass of an Escherichia coli protein is approximately 51 kDa. If the ribosome translates 20 nucleotides per second, according to commonly used estimation methods in biochemistry, the time required for a cell to synthesize this protein is 1 minute and 15 seconds.

Proposition 4:

A synthetic polyribonucleotide consisting solely of A and C in a 1:5 ratio is used as messenger RNA. In the amino acid composition of the protein synthesized by this RNA, the first five amino acids, ordered from smallest to largest molecular weight, exhibit a composition ratio of 150:30:5:5:1.

A. 1

B. 2

C. 3

D. 4

E. 1, 2

F. 1, 3

G. 1, 4

H. 2, 3

I. 2, 4

J. 3, 4

K. 1, 2, 3

L. 2, 3, 4

M. 1, 2, 4

N. 1, 3, 4

O. 1, 2, 3, 4

P. None of the above options is correct

Answer

Correct Answer: **K**

Detailed Explanation

Based on the provided information, analyze these four propositions.

Analysis of Proposition 1: Among the 64 mRNA codons, there are 3 stop codons (UAA, UAG, UGA), 1 start codon (AUG, encoding methionine), and the remaining codons encode 20 amino acids. Therefore, the total number of codons with or additionally possessing special signal functions is 4.

CHECKPOINT

0.5 PTS

Proposition 1: The total number of codons with or additionally possessing special signal functions in mRNA codons is 4

In mRNA codons, codon degeneracy refers to the number of codons encoding the same amino acid. The highest degeneracy is 6, observed for:

- Leucine (Leu/L): 6 (UUA, UUG, CUU, CUC, CUA, CUG)
- Arginine (Arg/R): 6 (CGU, CGC, CGA, CGG, AGA, AGG)
- Serine (Ser/S): 6 (UCU, UCC, UCA, UCG, AGU, AGC)

Thus, the number of amino acid types with the highest codon degeneracy is 3.

CHECKPOINT

0.5 PTS

Proposition 1: The number of amino acid types with the highest codon degeneracy in mRNA codons is 3

Therefore, in Proposition 1, the total number of codons with or additionally possessing special signal functions (4) does not match the number of amino acid types with the highest codon degeneracy (3). Proposition 1 is incorrect.

CHECKPOINT

1 PTS

Proposition 1 is incorrect

Analysis of Proposition 2: During translation, AUG encodes methionine (M), and UCU encodes serine (S), both of which are correct.

CHECKPOINT

0.5 PTS

Proposition 2: AUG encodes methionine (M), UCU encodes serine (S)

However, when the ribosome encounters the stop codon UAA, under normal circumstances, this termination signal is recognized by release factor proteins, which cleave and release the synthesized polypeptide chain, thereby terminating translation. In a system lacking release factors, this termination step cannot be completed. More importantly, there is no leucine tRNA in the cell that can pair with the UAA stop codon (leucine codons are UUA/G or CUX). Therefore, the ribosome cannot recruit leucine tRNA to extend the peptide chain, nor can it release the existing methionine-serine (M-S) peptide chain. The final result is that the ribosome stalls at the UAA codon, and the entire translation process halts at this point, preventing the synthesis of the M-S-L polypeptide.

CHECKPOINT

1 PTS

The translation process stalls, and the M-S-L polypeptide is not synthesized. Proposition 2 is incorrect

Analysis of Proposition 4: The protein has a mass of 51 kDa, and the average mass of a single amino acid is approximately 110 Da. Therefore, the number of amino acids is $\frac{51000}{110} = 463.64 \approx 464$, and the number of nucleotides required in the mRNA region encoding this protein is $464 \times 3 = 1392$.

CHECKPOINT

1 PTS

Proposition 3: The number of nucleotides required in the mRNA region encoding this protein is 1392

Thus, synthesizing an E. coli protein with a molecular mass of approximately 51 kDa would take about $\frac{1392}{20} = 69.6 \approx 70$ seconds. Proposition 3 is incorrect.

CHECKPOINT

1 PTS

Synthesizing this protein takes approximately 1 minute and 10 seconds, not 1 minute and 15 seconds. Proposition 3 is incorrect

Analysis of Proposition 4: Since codons consist of three nucleotides, there are eight possible codons that can form on this messenger RNA. The amino acids encoded by these codons include lysine (encoded by AAA, molecular weight 146.19 g/mol), asparagine (AAC, 132.12 g/mol), threonine (ACA, ACC, 119.12 g/mol), glutamine (CAA, 146.14 g/mol), histidine (CAC, 155.16 g/mol), and proline (CCA, CCC, 115.13 g/mol). Their molecular weights, ordered from smallest to largest, are proline, threonine, asparagine, glutamine, lysine, histidine.

CHECKPOINT

1 PTS

Proposition 4: The molecular weights of the amino acids in the protein, ordered from smallest to largest, are proline, threonine, asparagine, glutamine, lysine, histidine

For the frequency of occurrence of each amino acid:

$$\text{- Proline: } P(\text{Pro}) = P(\text{CCA}) + P(\text{CCC}) = \left(\frac{5}{6} \times \frac{5}{6} \times \frac{1}{6}\right) + \left(\frac{5}{6} \times \frac{5}{6} \times \frac{5}{6}\right) = \frac{25+125}{216} = \frac{150}{216}$$

$$\text{- Threonine: } P(\text{Thr}) = P(\text{ACA}) + P(\text{ACC}) = \left(\frac{1}{6} \times \frac{5}{6} \times \frac{1}{6}\right) + \left(\frac{1}{6} \times \frac{5}{6} \times \frac{5}{6}\right) = \frac{5+25}{216} = \frac{30}{216}$$

$$\text{- Histidine: } P(\text{His}) = P(\text{CAC}) = \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} = \frac{25}{216}$$

$$\text{- Asparagine and glutamine: } P(\text{Asn}) = P(\text{AAC}) = \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} = \frac{5}{216};$$

$$P(\text{Gln}) = P(\text{CAA}) = \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} = \frac{5}{216}$$

$$\text{- Lysine: } P(\text{Lys}) = P(\text{AAA}) = \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{216}$$

Therefore, for the five amino acids with the smallest molecular weights, their frequency ratios are: 150 (proline) : 30 (threonine) : 5 (asparagine) : 5 (glutamine) : 1 (lysine). Thus, Proposition 4 is correct.

CHECKPOINT

1 PTS

The correct ratio is 150 (proline) : 30 (threonine) : 5 (asparagine) : 5 (glutamine) : 1 (lysine). Proposition 4 is correct

Consequently, select option **K**.